SmartEvolve AI: Blockchain Enabled Self-Evolving AI System for Personalized Learning and Career Guidance

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***ABSTRACT*—Learning platforms that update a student's proficiency after every attempt can be gamed by repeating the same assessment, and the credentials they issue are hard to verify independently. This paper presents SmartEvolve AI, a prototype built from four cooperating modules. A Proficiency Module updates each skill with a learning rate that falls with the attempt count, a Prerequisite Resolver orders learning resources with a directed acyclic graph, a Career Matcher scores readiness for a target role, and a Credential Module anchors validated achievements on an Ethereum Virtual Machine (EVM) ledger as hashes, so that no cleartext personal data is written on-chain. Each module was tested on its own simulated data: 2,000 learners per seed over 10 seeds, synthetic curriculum graphs of up to 1,000 nodes, and the five methods of a smart contract. With ten retries of one assessment, the damped update cut score inflation from 38.9 to 23.8 points compared with a constant rate, but it tracked genuine improvement more slowly (mean absolute error 11.9 versus 6.8 points) and placed learners in the correct readiness state less often (79.6% versus 84.4%). A running mean was competitive on stationary skills. The benefit is therefore a trade-off, not a universal gain. No real students took part, so the combined system has not been tested with learners.**

***Index Terms*—adaptive learning, multi-module system, proficiency estimation, prerequisite graph, career readiness, blockchain credentials, smart contracts, educational data mining**

I. Introduction

Most learning platforms present a fixed sequence of courses, whatever the learner's starting point or pace. Platforms that do adapt must decide how far each new assessment score should move the stored estimate of a skill. If the estimate moves too slowly, genuine progress is ignored. If it moves too quickly, a learner can raise it by retaking the same quiz until the answers are memorized. Credentials add a second problem: certificates sit in separate databases, so an employer cannot easily tell whether a record has been altered.

Cognitive tutors and knowledge tracing model what a learner knows [1], [2], prerequisite graphs order content [3], and blockchain has been proposed for educational records [6], [7], but these pieces are usually built separately. A platform that estimates skills does not issue verifiable credentials, and a credential registry does not know how reliable the underlying scores are.

Here the pieces are combined into one pipeline of cooperating modules. The system contains four of them: a Proficiency Module, a Prerequisite Resolver, a Career Matcher, and a Credential Module. The paper makes four contributions. The first is a modular design around an attempt-damped proficiency update, with a stated bound: the estimate always moves toward the score and never past it. The second is a simulation study of that update against two simpler rules in three scenarios (stationary skill, genuine improvement, and repeated retries), including its effect on readiness-state decisions. The third is a prerequisite sequencer with cycle detection and an explainable career-readiness measure with a worked example. The fourth is a dual-hash credential scheme, together with an account of what it does not guarantee. The rest of the paper is organised as follows: related work comes first, then the data and the design of each module, then the experiments and their discussion, and finally the conclusions.

II. Related Work

Cognitive tutors established model-based adaptation of instruction [1]. Bayesian Knowledge Tracing and Deep Knowledge Tracing [2] estimate mastery from response sequences. Neural models can be accurate but are hard to interpret. Lightweight parametric rules with confidence terms are easier to audit and cheap to run, which motivates the update studied here. Prerequisites form a directed acyclic graph (DAG), and a valid study order is a topological ordering [3], [4]. Adaptive hypermedia systems have used such structures to guide learners [5].

Blockchain has been proposed for educational records and credentials [6]–[8], and Ethereum smart contracts [9] allow a public registry without a central verifier. Placing identifying data on a public ledger raises privacy concerns [10], so designs that store only commitments are preferred. Two questions received little attention in the work we reviewed: how a proficiency update behaves when a learner repeats an assessment to inflate the estimate, and what that does to downstream decisions such as readiness scoring. Both are

TABLE I
DATASETS AND WORKLOADS USED IN THE EXPERIMENTS

Workload	Size	Settings
Simulated learners	2,000 per seed, 10 seeds	Scenarios A, B, C; θ and noise as in Section III-A
Readiness test	2,000 learners per seed	5 skills, 5 assessments each, true levels $U(20, 95)$
Career profile	1 role, 5 skills	Data Analyst benchmark (R_i, W_i)
Curriculum DAGs	10–1,000 nodes	Synthetic prerequisite edges
Credential contract	5 methods	Solidity 0.8.20, Hardhat test network

examined here, together with the way validated results can be anchored on a ledger without exposing personal data.

III. Materials and Methods

A. Datasets and Workloads

Because no data from real students were available, every module was tested on simulated or synthetic workloads (Table I). The simulated learners have a true skill θ, and each assessment returns a noisy score around it. *Scenario A (stationary)* keeps θ fixed and varies only the noise. *Scenario B (improving)* raises θ by 4 points per attempt, as for a learner who is genuinely progressing. *Scenario C (retries)* has the learner retake the same assessment ten times, with the expected score rising by 6 points per retry through memorization, which models score gaming. Scores are clipped to $[0, 100]$, and the noise is $\mathcal{N}(0, 12^2)$ in A and B and $\mathcal{N}(0, 6^2)$ in C. The scenario parameters are assumptions, not estimates from real learners.

B. System Architecture

Fig. 1 shows the design. Each module has its own input, a decision step, and an output written to a shared record store (PostgreSQL with 19 tables and row-level security). The modules do not call each other directly, so any of them can be retrained or replaced without affecting the others. The backend is a Spring Boot service, the client is a React Native application, and the Credential Module talks to an EVM network through Web3j. Table II summarises the roles of the modules and Algorithm 1 gives the workflow that runs after each completed assessment. The system also delivers lessons, videos, and quizzes inside the application and tracks completion percentages, and a career knowledge base of more than 15 benchmark roles supplies the targets for the Career Matcher.

C. Proficiency Module

Let $P_s \in [0, 100]$ be the proficiency of a student in skill s, $A_k \in [0, 100]$ the score of assessment k, $n \geq 1$ the cumulative attempt count for the skill, and $w_d \in \{0.25, 0.35, 0.45\}$ the weight for beginner, intermediate, and advanced difficulty. Define

$$\lambda(n) = \frac{1}{\sqrt{\max(1, n)}}, \qquad \eta(d, n) = w_d\, \lambda(n), \tag{1}$$

$$P_s^{\text{new}} = \min\Big(100, \max\big(0,\ P_s^{\text{prev}} + \eta(d, n)\,(A_k - P_s^{\text{prev}})\big)\Big). \tag{2}$$

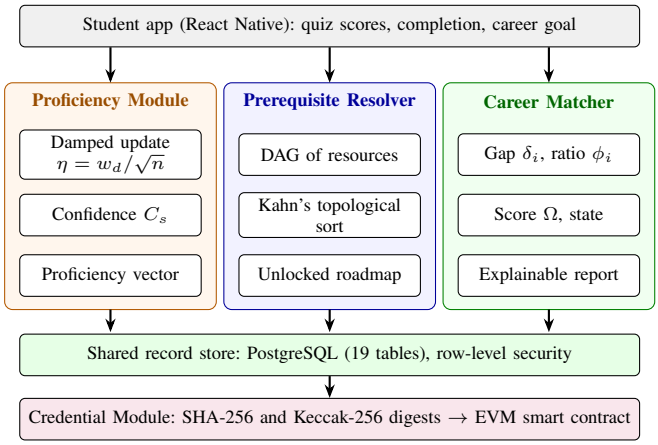


FIGURE 1: Four-module architecture. The Proficiency Module, Prerequisite Resolver, and Career Matcher write to a shared record store, and the Credential Module anchors validated achievements on-chain.

TABLE II
SUMMARY OF THE FOUR MODULES

Module	Input	Method	Output
Proficiency	Score, attempt count, difficulty	Damped update (Eq. 2)	Skill proficiency and confidence
Prerequisite Resolver	Resources, prerequisite edges	Kahn's topological sort	Ordered, unlocked roadmap
Career Matcher	Proficiency vector, role benchmark	Weighted fulfilment ratio (Eq. 5)	Ω, readiness state, deficits
Credential	Milestone score ≥ 80	SHA-256, Keccak-256 digests	On-chain credential record

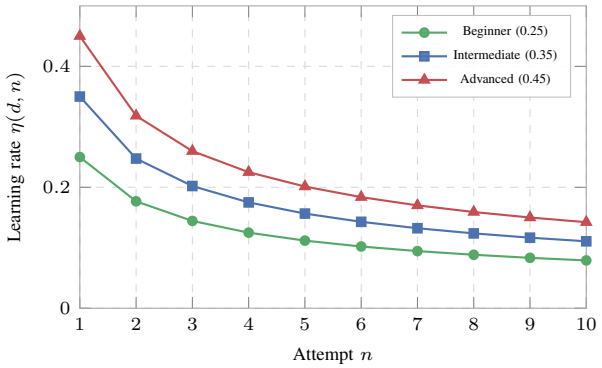
ALGORITHM 1 Workflow after each completed assessment

```
for each completed assessment (skill s, score A_k) do
    Proficiency Module updates P_s and C_s with Eqs. (2)–(3)
    Career Matcher recomputes Ω and the readiness state with Eq. (5)
    Prerequisite Resolver orders the resources with Kahn's method and unlocks those whose prerequisites are met
    if A_k ≥ 80 then
        Credential Module computes H_student, H_ach (Eqs. (6)–(7)) and submits them to the contract
    end if
end for
return updated roadmap and credentials
```

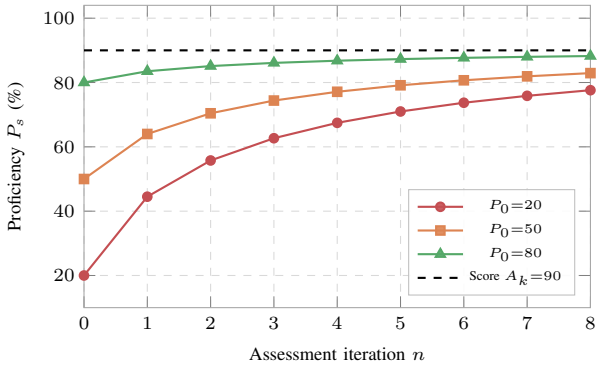
A confidence coefficient $C_s \in [0, 1]$ grows with evidence:

$$C_s^{\text{new}} = \min\big(0.99,\ C_s^{\text{prev}} + 0.08\,\lambda(n)\big). \tag{3}$$

Because $\eta \leq 0.45 < 1$, each update is a convex combination of P_s^{prev} and A_k, so P_s moves toward A_k and cannot exceed it when A_k is constant. Because η decreases with n, later attempts move the estimate less, which limits inflation from rapid retries but also slows the response to real improvement. Fig. 2(a) shows how $\eta(d, n)$ falls with the attempt count for the three difficulty tiers, and Fig. 2(b) shows the trajectories computed from Eq. (2) for three starting values when every score is 90 and $w_d = 0.35$.



(a) Effective learning rate by difficulty tier



(b) Proficiency trajectories for $A_k = 90$ ($w_d = 0.35$)

FIGURE 2: Behavior of the attempt-damped update.

TABLE III
CAREER COMPATIBILITY EXAMPLE (TARGET: DATA ANALYST)

Skill	P_i	R_i	W_i	δ_i	ϕ_i	$100\phi_i W_i$
SQL	45	80	0.25	35	0.563	14.06
Python	85	70	0.20	0	1.000	20.00
Statistics	35	75	0.25	40	0.467	11.67
Power BI	20	70	0.15	50	0.286	4.29
Visualization	30	65	0.15	35	0.462	6.92
Career compatibility Ω (Developing)						56.94

D. Prerequisite Resolver and Career Matcher

Resources $V = \{r_1, \ldots, r_m\}$ and prerequisites form a DAG $G = (V, E)$, where $(u, v) \in E$ means u must be completed before v is unlocked. Kahn's method [3] orders the resources in $\mathcal{O}(|V| + |E|)$ time: it repeatedly removes a resource whose prerequisites are all met, appends it to the path, and lowers the in-degree of its successors. If resources remain but none is free, the graph contains a cycle and a `CircularPrerequisiteException` is raised.

For a career requiring M skills with benchmark levels R_i and importance weights W_i ($\sum_i W_i = 1$), and student proficiencies P_i, the Career Matcher computes

$$\delta_i = \max(0,\, R_i - P_i), \qquad \phi_i = \min\Big(1, \frac{P_i}{R_i}\Big), \tag{4}$$

$$\Omega = 100 \sum_{i=1}^{M} \phi_i W_i. \tag{5}$$

Skills are reported as matched ($\delta_i = 0$), partial, or missing, with their deficits. The readiness state is *Ready* for $\Omega \geq 85$, *High potential* for $70 \leq \Omega < 85$, *Developing* for $50 \leq \Omega < 70$, and *Early stage* below 50. Table III applies the formulas to a student targeting a Data Analyst profile, whose initial profile gives $\Omega = 56.9$ (Developing). The last column of the table shows each skill's contribution to Ω; the remaining $100 - \Omega = 43.1$ points are the unmet share that the roadmap should close.

E. Credential Module

A credential is minted when a milestone assessment score is at least 80%. The backend computes two digests:

$$H_{\text{student}} = \text{Keccak-256}(\text{UUID}_{\text{student}} \,\|\, \text{Salt}), \tag{6}$$

$$H_{\text{ach}} = \text{SHA-256}(\text{CredentialUUID} \,\|\, \text{Title} \,\|\, P_{\text{final}} \,\|\, \text{Timestamp}). \tag{7}$$

The call `issueCredential` stores the credential UUID, H_{student}, the title, and H_{ach} on-chain; `setAuthorizedIssuer`, `verifyCredential`, and `revokeCredential` manage issuers, verification, and revocation. To verify, a third party recomputes H'_{ach} from the transcript supplied by the student and calls `verifyCredential`, which succeeds only if $H'_{\text{ach}} = H_{\text{ach}}$ and the credential is not revoked.

F. Baseline Update Rules

We compare the damped rule with two simpler rules applied to the same scores. The *Constant* rule uses a fixed rate $\eta = 0.35$, so every attempt moves the estimate by the same fraction of the error. The *Running mean* rule uses $\eta = 1/n$, which weights all attempts equally and is the most strongly damped of the three. The proposed rule, $\eta = 0.35/\sqrt{n}$, lies between them.

G. Evaluation Protocol

Every simulated value comes from a single script with fixed random seeds (1000–1009 for the main comparison). With N simulated learners, we report

$$\text{RMSE}_n = \sqrt{\tfrac{1}{N}\textstyle\sum_i \big(P_i^{(n)} - \theta_i\big)^2}, \tag{8}$$

$$\text{MAE} = \tfrac{1}{10}\textstyle\sum_{k=1}^{10} \tfrac{1}{N}\sum_i \big|P_i^{(k)} - \theta_i^{(k)}\big|, \tag{9}$$

$$\text{Infl}_r = \tfrac{1}{N}\textstyle\sum_i \big(P_i^{(r)} - \theta_i\big). \tag{10}$$

RMSE is used for Scenario A, MAE for Scenario B, and inflation for Scenario C. For the readiness test, each simulated learner has five skills with true levels drawn from $U(20, 95)$, takes five assessments per skill, and is assigned a readiness state from the estimated profile; this is compared with the state computed from the true levels, and agreement is reported as accuracy. Each metric is computed per seed, and we report the

TABLE IV
UPDATE-RULE COMPARISON (MEAN ± 95% CI OVER 10 SEEDS; LOWER IS BETTER; BEST IN BOLD)

Metric (points)	Damped	Constant	Running mean
Stationary RMSE, n=5	7.38±0.09	6.10±0.06	**5.32±0.07**
Stationary RMSE, n=10	4.60±0.09	5.51±0.08	**3.75±0.06**
Improving MAE (mean over 10)	11.92±0.05	**6.79±0.05**	10.85±0.06
Retry inflation, r=5	**8.91±0.02**	14.84±0.02	11.98±0.02
Retry inflation, r=10	**23.80±0.03**	38.94±0.07	25.43±0.03

mean of the 10 seed values with a 95% Student-t half-width (nine degrees of freedom). Because every seed uses 2,000 learners, the intervals are narrow and reflect Monte Carlo variability only. They do not describe uncertainty about real learners.

IV. RESULTS AND DISCUSSION

A. Proficiency Module

Table IV and Fig. 3(a) give the main comparison. Under repeated retries (Scenario C), the damped rule has the smallest inflation at both checkpoints: 8.91 points after five retries and 23.80 after ten, against 14.84 and 38.94 for the constant rate, a reduction of about 40% and 39%. The running mean is close at the tenth retry (25.43) but inflates more early on (11.98 at $r = 5$). Fig. 3(b) plots the mean inflation after each retry for 20,000 simulated learners. When the skill genuinely improves (Scenario B), the ordering reverses: the constant rate tracks best (MAE 6.79), and the damped rule lags most (11.92), even behind the running mean (10.85). On a stationary skill (Scenario A), the running mean has the lowest error at both $n = 5$ and $n = 10$. The damped rule is worse than the constant rate at $n = 5$ (7.38 versus 6.10) and better at $n = 10$ (4.60 versus 5.51), because its smaller late steps average out noise.

B. Career Matcher and Readiness States

Starting from the profile in Table III, we repeated one assessment per skill with score $A_k = 85$ and $w_d = 0.35$ in each iteration, updating proficiency with Eq. (2) and recomputing Ω with Eq. (5). Fig. 4(b) shows Ω rising from 56.9 (Developing) to 76.5 after one iteration (High potential) and passing the Ready threshold after the second (85.4). It saturates near 99 because ϕ_i is capped at 1 once a skill reaches its benchmark; the saturation reflects the capped metric and not mastery.

The damping also affects readiness decisions. After five assessments per skill, the estimated state matched the true state for 79.6% of 20,000 simulated learners with the damped rule (±0.8), against 84.4% (±0.9) for the constant rate and 87.1% (±0.5) for the running mean. Fig. 4(a) shows the confusion matrix of the damped rule. Every error is to an adjacent state, but the estimates lean low for strong learners: 36.4% of Ready learners were placed in High potential, because the damped estimate has not yet caught up with their true level. Early-stage learners are rare in this sample (320 of 20,000), and most of them were placed in Developing.

TABLE V
EFFECT OF THE BASE WEIGHT ON THE DAMPED RULE (MEAN ± 95% CI OVER 10 SEEDS; LOWER IS BETTER; BEST IN BOLD)

Setting	RMSE (n=10)	Improv. MAE	Inflation (r=10)
$w_d = 0.25$	6.56±0.10	13.99±0.04	**18.84±0.02**
$w_d = 0.35$ (default)	4.60±0.09	11.92±0.05	23.80±0.03
$w_d = 0.45$	**4.01±0.08**	**10.34±0.06**	27.75±0.04

TABLE VI
TOPOLOGICAL SEQUENCING BENCHMARK

$\|V\|$	$\|E\|$	Latency (ms)	Memory (MB)
10	18	0.42 ± 0.05	1.2
50	112	1.84 ± 0.12	2.8
100	280	3.65 ± 0.28	4.9
500	1450	9.72 ± 0.61	14.1
1000	3100	14.89 ± 0.94	24.6

C. Sensitivity to Noise and Memorization

Fig. 5 shows how the comparison changes with the scenario parameters. In Scenario A the damped rule is slightly worse than the constant rate when scores are accurate (RMSE 3.39 versus 1.87 at $\sigma = 4$) and better once the noise is large (4.62 versus 5.52 at $\sigma = 12$ and 6.37 versus 8.90 at $\sigma = 20$), with the crossover between $\sigma = 8$ and $\sigma = 12$. The running mean has the lowest error at every noise level. In Scenario C the inflation grows with the memorization gain for all three rules, and the damped rule stays lowest at every gain tested: for example 8.47 versus 14.35 points at 2 points per retry and 31.21 versus 46.45 at 10 points per retry.

Table V varies the base weight w_d of the damped rule. A smaller weight lowers the retry inflation (18.84 points at $w_d = 0.25$) but tracks genuine improvement more poorly (13.99), and a larger weight does the opposite (27.75 and 10.34). No setting is best on all three metrics, which suggests that the trade-off comes from the damping itself and not from the particular weight. The default of 0.35 is the intermediate difficulty weight and was not tuned. On a stationary skill, the running mean was more accurate than the damped rule at both checkpoints, and on genuine improvement the damped rule was worse than both baselines. The damped rule is therefore not a general improvement over simpler updates; its advantage is limited to settings where repeated retries are a real concern.

D. Prerequisite Resolver

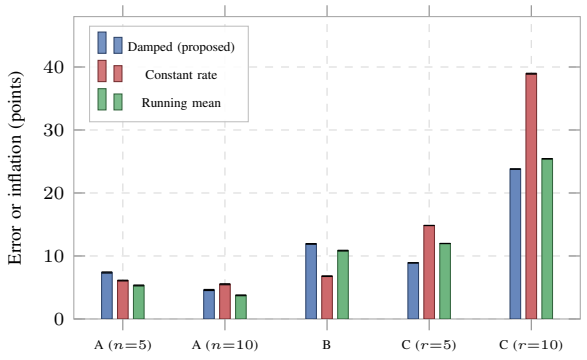
The sequencer was benchmarked on synthetic curriculum DAGs of 10 to 1,000 nodes (Table VI). Latency grows roughly in line with graph size, as expected from the $\mathcal{O}(|V| + |E|)$ bound.

E. Credential Module

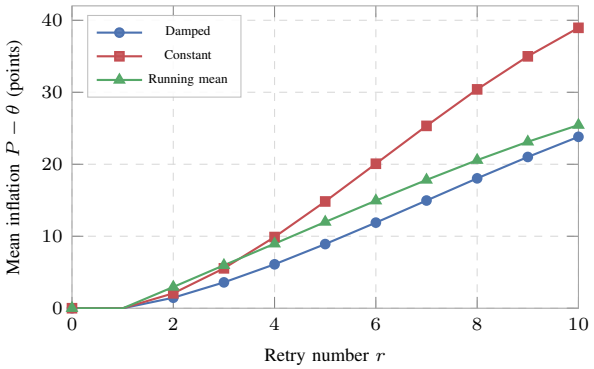
Table VII reports gas for each contract method. The ETH cost is gas × 15 Gwei.

F. Comparison with Prior Findings

Our results agree with the literature on one point: the choice of model or rule depends on the setting. Knowledge-tracing studies report that no single model is best on every dataset [2], and here the best update rule changed with the scenario. In

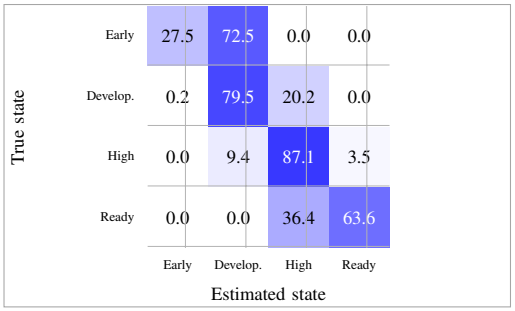


(a) Error or inflation by scenario (mean ± 95% CI)

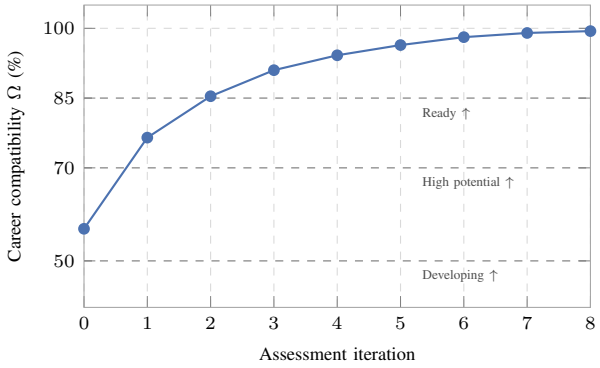


(b) Inflation under ten retries of one assessment

FIGURE 3: Comparison of the three update rules.

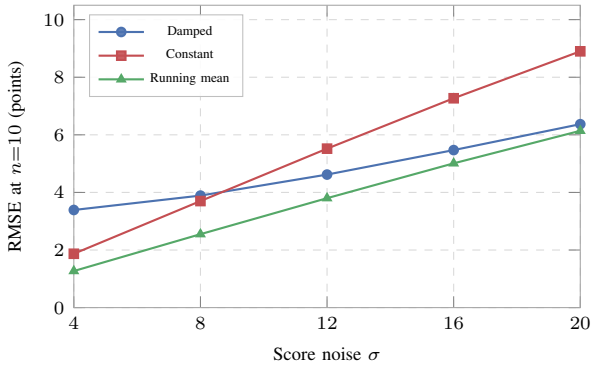


(a) Readiness-state confusion, damped rule (row %)

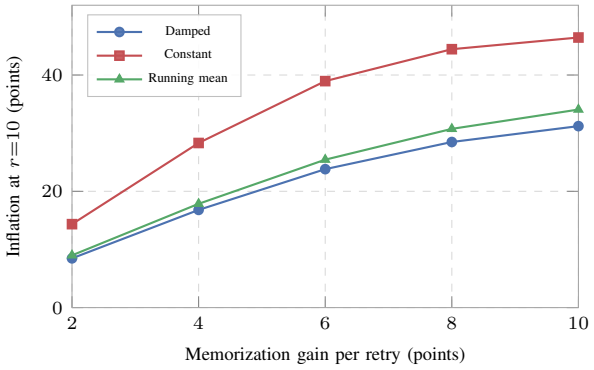


(b) Career compatibility Ω over repeated assessments

FIGURE 4: Career readiness: state agreement and evolution.



(a) Stationary skill: error against noise level



(b) Retries: inflation against memorization gain

FIGURE 5: Sensitivity of the update rules to noise and memorization.

capability, the system differs from static platforms (no adaptive proficiency and no career-gap analysis), from knowledge-tracing systems (adaptive, but without a prerequisite roadmap or credentials), and from blockchain credential systems (tamper-evident records, but no adaptive profile). These are typical capabilities and not a benchmark. Accuracies quoted in the knowledge-tracing and blockchain-education literature rest on other data and evaluation schemes, so our numbers are not ranked against them.

G. Limitations

First, all learners are simulated, and the score model, noise levels, and memorization rate are assumptions; the conclusions about inflation and tracking need confirmation

TABLE VII
GAS CONSUMPTION OF CONTRACT METHODS

Method	Gas	ETH @15 Gwei
Deployment	682,419	0.01024
`setAuthorizedIssuer`	28,142	0.00042
`issueCredential`	72,410	0.00109
`revokeCredential`	31,210	0.00047
`verifyCredential`	28,150	0 (off-chain read)

on real assessment data. Second, the difficulty weights and the square-root form were not fitted to real data. Third, the sequencing and gas benchmarks use synthetic workloads and a test network, and the gas values must be re-measured before publication. Fourth, the career benchmarks are a controlled set of roles that need periodic review. Fifth, the modules were tested separately, and no pilot study with students was run, so the effect on learning outcomes is unknown.

On security and privacy, altering an off-chain record changes its hash, so verification fails and a compromised server cannot silently modify an issued credential. Verification shows that a record is unchanged, not that its issuer was legitimate, so issuance is restricted to authorized accounts. Only a salted student hash and an achievement digest are stored on-chain; names, e-mail addresses, and transcripts stay in the access-controlled database. Pseudonymous hashes may still count as personal data under regulations such as GDPR, and immutability conflicts with erasure rights, so the design reduces compliance obligations without removing them.

V. CONCLUSION

This paper presented a four-module prototype that links an attempt-damped proficiency update with prerequisite sequencing, career-readiness scoring, and hashed on-chain credentials, and each module was tested on its own simulated data. In simulation, damping reduced retry-driven inflation by roughly 39% relative to a constant rate, but it tracked genuine improvement less closely and placed learners in the correct readiness state less often (79.6% versus 84.4%). A running mean was competitive on stationary skills. The main lesson is about reporting: whether damping helps depends on how much gaming matters, and the trade-off should be stated with the result. Future work will fit the model parameters on pilot data, compare the update with knowledge-tracing baselines, test the modules together on a class-wide dataset in which scores, readiness, and credentials refer to the same students, and explore zero-knowledge proofs for threshold claims such as grades.

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